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Convergent validity assessment of formatively measured constructs in PLS-SEM

On using single-item versus multi-item measures in redundancy analyses

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Abstract

Purpose – Researchers often use partial least squares structural equation modeling (PLS-SEM) to estimate path models that include formatively specified constructs. Their validation requires running a redundancy analysis, which tests whether the formatively measured construct is highly correlated with an alternative measure of the same construct. Extending prior knowledge in the field, this paper aims to examine the conditions favoring the use of single vs multiple items to measure the criterion construct in redundancy analyses.

Design/methodology/approach – Merging the literatures from a variety of fields, such as management, marketing and psychometrics, we first provide a theoretical comparison of single-item and multi-item measurement and offer guidelines for designing and validating suitable single items. An empirical comparison in the context of hospitality management examines whether using a single item to measure the criterion variable yields sufficient degrees of convergent validity compared to using a multi-item measure.

Findings – The results of an empirical comparison in the context of hospitality management show that, when the sample size is small, a single item yields higher degrees of convergent validity than a reflective construct does. However, larger sample sizes favor the use of reflectively measured multi-item constructs, but the differences are marginal, thus supporting the use of a global single item in PLS-SEM-based redundancy analyses.

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Originality/value – This study is the first to research the efficacy of single-item versus multi-item measures in PLS-SEM-based redundancy analyses. The results illustrate that a convergent validity assessment of formatively measured constructs can be implemented without triggering a pronounced increase in survey length.

Keywords PLS-SEM, Partial least squares, Structural equation modeling, Convergent validity, Redundancy analysis, Formative measurement models

Paper type Research paper

Introduction

Structural equation modeling (SEM) has become the norm for analyzing cause–effect relationships between latent variables. When conducting business research, authors often desire to test complete theories and concepts. This is one of the major reasons why they have embraced SEM (Babin *et al.*, 2008). While SEM is a general term encompassing a variety of statistical models, covariance-based SEM (CB-SEM; Jöreskog, 1978) is the more widely used approach, with many researchers simply referring to CB-SEM as SEM. However, research has suggested alternative SEM techniques, most notably partial least squares structural equation modeling (PLS-SEM; Lohmöller, 1989; Wold, 1982), which has recently come to the attention of a variety of disciplines, such as hospitality management, human resource management, international business, marketing and tourism (Ali *et al.*, 2018a,b; do Valle and Assaker, 2016; Hair *et al.*, 2012b; Richter *et al.*, 2016; Ringle *et al.*, 2018). Reviews of PLS-SEM use in these fields show univocally that researchers often justify their choice of method on the grounds of their measurement model specification, because PLS-SEM allows for estimating formatively specified constructs without identification concerns (Sarstedt *et al.*, 2016b)[1].

Unlike their reflective counterparts, indicators in formative measurement models form the construct by means of linear combinations (Diamantopoulos *et al.*, 2008). For instance, a change in a respondent's assessment of the trait that an indicator is capturing alters the indicator's value, which, in turn, results in a change of the construct's value. Formative measurement has become important in business research (Bollen and Diamantopoulos, 2017; Diamantopoulos *et al.*, 2008), as it allows for isolating indicators' relative impact when forming a construct (i.e. in the context of a certain nomological net represented by the PLS path model), which makes more nuanced managerial recommendations possible (Albers, 2010).

The convergent validity assessment is a requirement for the empirical evaluation of formative measurement models in PLS-SEM. Convergent validity is the extent to which a measure relates to other measures of the same phenomenon (Hair *et al.*, 2017a, Chapter 5). A redundancy analysis, in which each formatively specified construct is correlated with an alternative measure of it, provides this assessment (Chin, 1998). The result allows for determining whether, jointly, the formative indicators represent the construct of interest adequately.

When running a redundancy analysis, researchers can opt for a global single-item, or for a reflectively specified multi-item measure, as the criterion variable. The two approaches to constructing measurement have apparent advantages over one another, with the literature debating these (Bergkvist and Rossiter, 2007; Fuchs and Diamantopoulos, 2009; Diamantopoulos *et al.*, 2012; Loo, 2002; Wanous *et al.*, 1997). However, prior research has not yet offered any recommendations on the use of single-item vs multi-item measures in the context of PLS-SEM-based redundancy analyses. For example, Hair *et al.* (2014, p. 112) noted that, in a redundancy analysis, "each formatively measured construct is correlated with an alternative reflective or single-item measurement of the same construct", without offering any perspective on a concrete choice. Similarly, textbooks (Ramayah *et al.*, 2017; Hair *et al.*, 2017a) and tutorial articles (Ali *et al.*, 2018b; Sarstedt *et al.*, 2014) that discuss the key issue of measurement model evaluation in PLS-SEM remain silent on this issue. Specifically, while Hair *et al.* (2017a, p. 141) stated that the use of a global single item may suffice in a redundancy analysis, where "the aim is not to fully capture the content domain of the construct but only to consider its salient elements", research has not empirically tested this claim. Prior studies that offer guidance for choosing between single and multiple items (Diamantopoulos *et al.*, 2012; Fuchs and Diamantopoulos, 2009; Loo, 2002; Nagy, 2002; Sloan *et al.*, 2002) univocally compare the measures' relative performance outside the specific context of the redundancy analysis, which assumes a specific threshold for establishing convergent validity. Furthermore, all these studies do not use an indicator weighting method such as PLS-SEM but rely on sum scores regressions. But even if a researcher opts for using a global single item, what should this item look like? While prior research has brought forward numerous procedures for developing multi-item scales (Carpenter *et al.*, 2016; DeVellis, 2016; Kuppelwieser *et al.*, 2018), no corresponding procedure has been brought forward in the context of single items (Diamantopoulos, 2006; Sarstedt *et al.*, 2016a). Considering these ambiguities, it is not surprising that many researchers fail to report the assessment of formative measurement models' convergent validity. For instance, Ali *et al.*'s (2018a) review of PLS-SEM use in hospitality management shows that none of the relevant studies ran a redundancy analysis.

Addressing this gap in research, this study examines the conditions favoring the use of each measure type when running a redundancy analysis in PLS-SEM. Merging the literatures from a variety of fields, such as management, marketing and psychometrics, we first offer a theoretical comparison of single-item and multi-item measurement. Addressing related calls in prior research (Diamantopoulos *et al.*, 2012), we then propose a guideline for designing and validating single-item measures, which combines aspects from the development of multi-item scales and prior research on single-item scaling. The results of an empirical comparison in the context of hospitality management show that using a single item to measure the criterion variable yields sufficient degrees of convergent validity and excels multi-item measurement in terms of out-of-sample prediction. While single items do not allow for capturing a construct's content domain in full, their use is beneficial in the context of a redundancy analysis. This result is robust for different sample sizes and numbers of indicators in the multi-item measure. As such our findings offer not only empirical support for corresponding claims in the prior literature (Hair *et al.*, 2017a; Ramayah *et al.*, 2017) but also important recommendations on how to run a redundancy analysis to assess formatively measured constructs in PLS-SEM.

Single versus multiple items in redundancy analyses

Formative measurement model evaluation in PLS-SEM follows a multi-step process, which involves the assessment of indicator collinearity, the indicator weights' significance and size and convergent validity. Convergent validity refers to the extent to which the scores of a formatively measured construct correlate with the scores of another construct representing the same concept. Its assessment requires running a redundancy analysis (Chin, 1998), in which the formatively measured construct serves as an exogenous latent variable linked to an endogenous latent variable (i.e. a criterion construct) that measures the same concept. The structural path linking the two constructs indicates the degree to which the set of formative indicators represents the construct of interest. According to Hair *et al.* (2017a, Chapter 5), the path coefficient should be 0.7 or higher, which means the formative construct should explain at least 50 per cent of the criterion construct's variance. More conservatively, Chin (2010, p. 683) suggested, "In general, we would expect a path of 0.80 or above to be suggestive of securing an adequate (i.e. comprehensive) set of formative measures assuming convergent validity (i.e. adequate loadings) for the reflective set. A path of 0.90 or above would indicate an extremely strong result".

In the original presentation of the redundancy analysis, Chin (1998) suggested a reflective multi-item measurement of the criterion construct. From a psychometric perspective, this approach seems favorable, as the combination of multiple items in reflective measurement averages out random error in the items, thereby increasing the reliability (Sarstedt and Wilczynski, 2009). Hence, with more items per construct, the average measurement error decreases, whereas the measurement accuracy increases (Peter, 1979). As multiple items cover a larger number of distinct construct facets, they also offer higher degrees of construct validity (Wanous *et al.*, 1997). Likewise, using multi-item measures gives researchers more degrees of freedom when handling missing values, as imputation procedures can use the information inherent in other items to identify reasonable values (Sloan *et al.*, 2002).

Nevertheless, reaping the benefits of multi-item measures comes at a price, because adding a set of reflective items that measures each formatively specified construct redundantly increases the survey length, leading to lower response rates and decreased data quality (e.g. due to mindless response behavior; Drolet and Morrison, 2001). This is particularly problematic in the context of a redundancy analysis, as formative measurement models generally already include more indicators than their reflective counterparts – as evidenced in several reviews of PLS-SEM use (Hair *et al.*, 2012a,b, Nitzl, 2016a). Hence, adding more items extends the already-pronounced survey length even more.

Hair *et al.* (2017a, p. 141) recommended using a global single item "that summarizes the essence of the construct the formative indicators purport to measure" as a solution. In the context of, for example, customers' service satisfaction with a restaurant, such a global single would involve asking: "Overall, how satisfied are you with the service of the restaurant?" Single items' simplicity and brevity make them stand out. For example, designing single items usually requires less effort than the construction of multi-item scales (Gardner *et al.*, 1998). Furthermore, single items reduce the cognitive demands required of the respondents, which increases the response rates (Drolet and Morrison, 2001) and reduces suspicious response patterns, such as straight-lining (i.e. respondents giving an identical response to all the items presented in a grid), or rote keystroke responses within question blocks (Fuchs and Diamantopoulos, 2009). Finally, single-item measures offer the advantage of straightforward and flexible adjustment to new research situations and contexts. For example, when surveying a new construct dimension, the researcher must construct a new multi-item scale in a multi-level process, whereas single items can be easily adjusted to meet the altered research objective (Nagy, 2002).

Even though they offer several practical advantages, single items normally lag behind their multi-item counterparts in terms of psychometric properties. For example, based on their large-scale simulation study in which the authors compared the predictive validity of single-item and multi-item measures under a variety of measurement and data settings, Diamantopoulos *et al.* (2012, p. 446) concluded that "opting for SI [single-item] measures in most empirical settings is a risky decision as the set of circumstances that would favor their use is unlikely to be frequently encountered in practice". Sarstedt *et al.* (2016b) recently confirmed these results, additionally illustrating the problems that come with selecting a suitable single item from an existing scale.

However, in a redundancy analysis, the aim is *not* to validly measure the concept in full, but rather to cover the core of the concept's content domain (Hair *et al.*, 2017a, Chapter 5). By using a global single item, researchers assume that respondents "automatically consider different aspects of the construct" (Fuchs and Diamantopoulos, 2009, p. 204). That is, a global single item "allows the subject to take personally salient features of the situation into account when providing a response" (Youngblut and Casper, 1993, p. 459). Respondents therefore ignore aspects that are not relevant for their situations, and differentially weight the aspects to provide a single rating (De Boer *et al.*, 2004). This particularly holds for highly complex constructs, whose dimensions may not be adequately captured by multiple items. As a result, Scarpello and Campbell (1983), for example, contended that global single items should be preferred when measuring attributes such as satisfaction.

To summarize, compared with corresponding multi-item measure, a global single item may not yield the same but a *sufficient* degree of predictive validity (Sarstedt *et al.*, 2016a) in the context of a redundancy analysis, while offering manifold practical advantages. Our empirical comparison sets out to test this notion.

Empirical comparison

Study design and data

Our empirical comparison of single-item and multi-item measures in the context of a redundancy analysis draws on data from Ruiz *et al.*'s (2008) service value index model, which measures consumers' overall assessment of the utility of a service based on perceptions of what is received and what is given. Unlike prior research, which univocally draws on a reflective measurement model specification, these authors conceptualize service value formatively. Specifically, Ruiz *et al.* (2008) considered service value as a higher-order construct with four lower-order components that jointly form the higher-order component service value. In line with this conceptualization, our analysis applies the two-stage approach (Hair *et al.*, 2018, Chapter 2) to derive the latent variable scores of the four reflectively measured lower-order components' confidence benefits (CB), perceived sacrifice (PS), service equity (SE) and service quality (SQ) as formative indicators of a single service value construct.

On these theoretical grounds, we determine the construct's items and follow established procedures for conducting surveys (i.e. design of the questionnaire, pre-test and survey improvement, final data collection and post hoc test; Fink, 2017). We adapted seven items from Grewal *et al.*'s study (1998), as well as from Sweeney and Soutar's study (2001), which primarily relate to service value's functional aspect, to measure the multi-item criterion construct. In addition, we developed a global single item, designed to allow a holistic assessment of service value (see the next section). Finally, drawing on Taylor and Baker (1994) and Zeithaml, *et al.* (1996), we queried each respondent's service satisfaction and repurchase intention. As we collected the data in Malaysia, we used the translation and back-translation procedure (Brislin, 1970) to ensure that the questionnaire items and response formats had equivalent meanings compared to the original versions. All the items were measured on five-point Likert scales. Table AI offers an overview of all the construct measures and Table AII shows the item-specific descriptive statistics.

We conducted a pre-test of the instrument on three college students and five adult volunteers from the target population. The purpose was to elicit comments and suggestions regarding potential flaws in terms of the questionnaire format, design and wording that we adapted from Ruiz *et al.*'s (2008) study. In addition, this pre-test allowed us to address issues related to response bias and miscomprehension prior to the data collection (Fink, 2017). We subsequently pilot tested the revised questionnaire on another 30 respondents from the target population to identify errors and to optimize the survey design.

Using a mall intercept approach in one of Kuala Lumpur's largest malls (Velu and Naidu, 2009), we collected data from consumers who regularly dine (i.e. at least once a week) at one of the five major fast food restaurants in Malaysia (McDonald's, KFC, Subway, Burger King and Pizza Hut). We asked the respondents to answer the survey by keeping in mind the restaurant they mostly visit. Our data collection took place in October 2016 and produced 500 questionnaires with complete information (we discarded ten questionnaires due to nonresponse). The respondents were primarily ethnic Chinese, females, between 21 and 30 years old with a high school degree and a monthly income of RM2,000 or less (Table AIII shows a detailed breakdown of the sample characteristics). To test for non-response bias, we compared the first and last waves of respondents on all the variables (Armstrong and Overton, 1977) but found no statistically significant differences ($p < 0.01$). Hence, we conclude that there is no critical degree of non-response bias.

A post hoc power analysis (Fink, 2017), assuming an effect size of 0.15 and a power level of 80 per cent, indicated a minimum sample size of 85. We therefore broke down the sample randomly into five sets of 100 observations each, with each set still meeting the minimum sample size requirements. This step allowed us to examine the sample size effects in the redundancy analysis, as they could affect the path coefficient in the redundancy model through the endogenous construct's reliability. Specifically, Kopalle and Lehmann (1997) found that increasing a sample size decreases the scale reliability, which decreases the path coefficient estimate in the redundancy model (Diamantopoulos *et al.*, 2012)[2]. We compared the redundancy analysis results of the global single item and the multiple items in respect of 100, 200, 300, 400 and 500 observations.

Finally, following Diamantopoulos *et al.* (2012), we varied the number of items in the reflective multi-item measurement. According to classical test theory and the domain sampling model (Churchill, 1979), item measurement involves random error, which the combination of numerous items averages out. Hence, measurement error decreases with an increase in the number of items, whereas measurement accuracy increases (Ryan *et al.*, 1995). Reducing the number of items will therefore increase random error (thereby decreasing scale reliability) and decrease the path coefficient in the redundancy analysis.

Generation and validation of the global single item

To generate and validate the global single item of service value, we merged literatures from a variety of fields on scale development and comparisons of single- and multi-item measures (Diamantopoulos *et al.*, 2012; Loo, 2002; Nagy, 2002; Wanous and Hudy, 2001; Wanous and Reichers, 1996). Our procedure follows a three-step process, which covers the aspects of item generation, reliability, convergent and criterion validity assessment.

Step 1: Item generation. To develop a suitable global single item, we followed a deductive approach (Hinkin, 1998) that uses a theoretical definition of the service value concept to guide the item generation. Drawing on Zeithaml's (1988) definition of product value, we draw on Ruiz *et al.* (2008) who conceptualized service value as the consumers' overall assessment of what they receive from the exchange, and sacrifice, or what they give up. In line with this theoretical definition, we reviewed the measurement scales by Grewal *et al.* (1998) and Sweeney and Soutar (2001), which primarily relate to service value's functional aspects. Specifically, we sought to create measure that would tap into time, effort and money aspects of the service value. This assessment yielded the following item: "Overall, this fast-food restaurant provides a good service value". Before data collection, we assessed the item's face validity with three college students and five adult volunteers from the target population.

Step 2: Reliability assessment. To assess the single item's reliability, we draw on Wanous and Reichers (1996) and use a modified version of the well-known correction of attenuation formula:

$$r_{x'y'} = \frac{r_{xy}}{\sqrt{r_{xx} \cdot r_{yy}}}, \tag{1}$$

where r_{xy} is the measured correlation between construct x and y, r_{xx} (r_{yy}) is the reliability of construct x (y) and $r_{x'y'}$ is the estimated correlation between the two constructs corrected for attenuation. The formula is usually applied to compare two different constructs, but can also be used in situations where x and y represent single-item and a multi-item measures of the same concept, respectively. In this case, we can assume that $r_{x'y'} = 1$, which results in:

$$r_{xx} = \frac{r_{xy}^2}{r_{yy}}, \tag{2}$$

where r_{xx} is the reliability estimate of the single-item measure x. Hence, reliability assessment by means of this formula requires the simultaneous administration of a single-item and a multi-item measure of the same concept.

In the context of our study, r_{xy}^2 is the squared correlation between the composite score of SV1-SV7 and the global single item (SV8). Following prior research (Wanous *et al.*, 1997; Wanous and Hudy, 2001; Sarstedt and Wilczynski, 2009) and because our reliability assessment is model-independent, we use Cronbach's alpha as the reliability measure (r_{yy}). The results in Table I show that across all sample size constellations, the single item achieves a reliability value (r_{xx}) of 0.789 and higher. For example, for $N = 100$, the correlation between the single-item and multi-item measure of service value is 0.771 and the multi-item measure's reliability is 0.933, yielding a reliability estimate of the single item of 0.826. These results provide clear support for the reliability of the global single-item measure.

Step 3: Convergent and criterion validity assessment. Step 3 involves testing the convergent and criterion validity assessment of the global single item. Convergent validity is the degree to which the single item correlates positively with the reflective multi-item measure of service value. Similarly, criterion validity is the extent to which the global single item correlates with a criterion measure with which it is supposed to be related (Mooi *et al.*, 2018).

Table I shows the squared correlations between the composite scores of the seven reflective items and the global single item (r_{xy}^2). Taking the square root of these estimates yields the global single items' convergent validities (Christophersen and Konradt, 2011;

Sample size N	r^2_{xy}	Reliability r_{yy}	r_{xx}	Criterion validity Customer satisfaction	Repurchase intention
100	0.771	0.933	0.826	0.727	0.484
200	0.759	0.919	0.826	0.752	0.619
300	0.738	0.912	0.809	0.737	0.602
400	0.724	0.914	0.792	0.702	0.551
500	0.722	0.915	0.789	0.690	0.556

Notes: r_{xy} corresponds to the convergent validity estimate; all correlations are significant at 0.01

Table I.
Reliability and validity assessment

Gardner *et al.*, 1998; Loo, 2002). The results indicate that all the r_{xy} values are above 0.8, providing support for the global single item's convergent validity.

To assess the global single item's criterion validity, we used customer satisfaction and repurchase intention as two potential outcomes of service value (Tam, 2004). The results in Table I show that the global single item has significant and pronounced correlations with both criterion variables. For example, the global single item has correlations of around 0.7 with the multi-item measure of customer satisfaction across all sample sizes. The correlations with repurchase intentions are lower but still substantial, providing support for the single item's criterion validity.

Assessment of the measurement models

We used SmartPLS 3.2.7 (Ringle *et al.*, 2015) to estimate the path models, applying the path weighting scheme. We followed Hair *et al.* (2018, Chapter 2) to estimate the higher-order construct of the service value and applied the two-stage approach. Before assessing the convergent validity of the formative service value measure, we examined the lower-order components' reliability and validity (first stage), followed by the indicators' degree of collinearity, as well as the significance and relevance of the indicator weights derived in the second stage (Hair *et al.*, 2017a, Chapter 5). Furthermore, following standard evaluation guidelines, we assessed the reliability and validity of the reflective service value measure that serve as the criterion variable in our redundancy analysis (Hair *et al.*, 2012b).

The assessment of the lower-order components' measurement models shows that all the measures are reliable and valid. Except for the third PS item, all the loadings exceed the commonly used threshold value of 0.7 across all the sample sizes. Likewise, the average variance extracted, which indicates the amount of variance that the construct captures in relation to the variance due to measurement error, is higher than the 0.5 threshold for all constructs. Similarly, all the composite reliability values are above 0.7, supporting the measures' internal consistency reliability. Discriminant validity assessment using Henseler *et al.*'s (2015) heterotrait–monotrait ratio of correlations (HTMT) measure shows that, across all the sample sizes, the HTMT values are below the more conservative threshold of 0.85 and significantly lower than 0.9, thus supporting the measures' discriminant validity (Franke and Sarstedt, 2018).

Similarly, our measurement model assessment provides support for the reliability and validity of the reflective service value criterion construct. All the loadings exceed the commonly used threshold value of 0.7 across all the sample sizes. Likewise, the average variance extracted is higher than the critical value of 0.5. All the composite reliability values are around 0.9, supporting the measures' internal consistency reliability.

All the formative indicators (i.e. the lower-order components) across all the sample sizes yield VIF values below the threshold value of 3.3 (Diamantopoulos and Siguaw, 2006). Hence, collinearity is not at a critical level in any of the sample size constellations. We ran bootstrapping with 10,000 subsamples and used the no sign changes option (Streukens and Leroi-Werelds, 2016) to test the indicator weights' significance based on the 99 per cent bias-corrected and accelerated (BCa) confidence interval (Hair *et al.*, 2017a, Chapter 5). The results show that all the indicator weights are significant ($p < 0.01$), except for SE and SQ at a sample size of 100. However, as the corresponding indicator loadings are significant and above the 0.5 threshold (SE: loading = 0.530, $p < 0.01$; SQ: loadings = 0.771, $p < 0.01$), these indicators are not discarded (Hair *et al.*, 2017a, Chapter 5). Comparing the indicator weights shows that CB and PS have the strongest impact on service value, followed by SE and SQ.

Redundancy analyses

To carry out the redundancy analyses, we established separate models in which the formative measurement of the service value predicts:

- the global single-item measure; and
- the reflective multi-item measure of service value[3].

Furthermore, we varied the sample size across five conditions (100, 200, 300, 400 and 500) and the number of items in the reflective measurement model between two and seven items. To do so, we subsequently deleted one item with the lowest loadings from the measurement model and re-estimated the redundancy model for each condition.

The results in Table II show that the global single-item measure produces path coefficients of 0.7 and higher for sample sizes (N) of up to 300. In higher sample sizes of 400 and 500, the path coefficients in the redundancy model drop to, respectively, 0.670 and 0.632. The use of a multi-item measure produces a more disperse result. The redundancy analysis yields path coefficients above 0.7 for $N = 100$, regardless of the number of indicators.

Increasing the sample size of a certain number of indicators has a negative impact on the path coefficient estimates. For example, when increasing the sample size from 100 to 200 in the three-indicator measurement, the path coefficient estimate drops from 0.738 to 0.680. In contrast, increasing the number of indicators of larger sample sizes offsets this negative effect partially, but only to a certain degree. Only the seven-item model meets the 0.7 threshold for $N = 400$ and neither model achieves a path coefficient estimate above 0.7 for $N = 500$. The path coefficient estimate of the multi-item model is higher at these sample sizes than that of the global single item of four or more indicators, supporting prior research on the predictive validity of single items (Diamantopoulos *et al.*, 2012; Sarstedt *et al.*, 2016b).

The above analyses assess the measures convergent validity taking the entire data into account. Complementing this in-sample prediction perspective, we ran Shmueli *et al.*'s (2016) PLSpredict procedure to generate holdout sample-based point predictions on a construct level (Sarstedt *et al.*, 2017). Applying PLSpredict on all the models using ten folds and ten repetitions, we found that that the single-item measure of service value produces equal or lower RMSE values across all combinations of sample size and number of indicators. For example, for $N = 500$ and seven indicators, the RMSE of the single-item model is 0.553 compared to 0.556 for the multi-item model. This divergence increases to 0.569 versus 0.597 for $N = 100$ and seven indicators. These results suggest that the single item is not only sufficient in terms of (in-sample) convergent validity but also excels the multi-item measure in terms of out-of-sample prediction.

	Path coefficient estimates						
Sample size	Global single-item measure	Reflective multi-item measure					
N	1	2	3	4	5	6	7
100	*0.776*	*0.752*	*0.738*	*0.736*	*0.740*	*0.740*	*0.755*
200	*0.750*	0.672	0.680	*0.700*	*0.714*	*0.729*	*0.730*
300	*0.700*	0.642	0.650	0.683	0.690	*0.700*	*0.701*
400	0.670	0.630	0.633	0.680	0.680	0.690	*0.700*
500	0.632	0.624	0.623	0.660	0.664	0.681	0.690

Notes: We reduced the number of indicators in the multi-item measurement from seven to two, based on the lowest item-to-total correlation. Italics indicate values that meet the threshold for redundancy analyses

Table II.
Redundancy analysis results

Mediation analyses

In a final step, we analyzed whether the single-item measure explains a significant share of the relationship between the formative and reflective measure of service value by running a mediation analysis. Specifically, following Nitzl (2016b), we established a series of path models in which the single-item measure serves as the mediator in the relationship between the formative and reflective measures of service value. We analyzed each model using bootstrapping with 10,000 subsamples and the no sign changes option (Streukens and Leroi-Werelds, 2016) to determine the significance of the direct and indirect effects based on the 99 per cent BCa confidence interval (Hair *et al.*, 2017a, Chapter 5). In addition, we analyzed the variance accounted for (VAF). Technically, the VAF represents the ratio of the indirect effect to the total effect in the mediator model. Thereby, we can determine the extent to which the variance of the dependent variable is directly explained by the independent variable. In other words, the VAF expresses how much of the target construct's variance is explained by the indirect relationship via the mediator variable.

Across all combinations of sample sizes and number of indicators, we find significant ($p < 0.01$) direct effects between the formative and reflective service value measures, as well as significant indirect effects ($p < 0.01$) via the single-item measure. These results suggest that the single item partially mediates the relationship between the formative and reflective measures (Zhao *et al.*, 2010). The VAF values in Table III show that the single item accounts for a significant share of the reflective measures' variance. All VAF values are around 0.5, suggesting that the single item acts as a potential substitute for the multi-item measure as it explains a significant share of its variance in the mediation model.

Conclusion and discussion

Our theoretical discussion and empirical comparison clearly show that, in the context of a redundancy analysis, the use of single items to measure the criterion construct is clearly sufficient. While single items yield lower levels of predictive validity, the differences are generally marginal and therefore of little relevance for the assessment of convergent validity[4]. In fact, for sample sizes commonly encountered in the PLS-SEM applications, researchers would need a high number of items for the multi-item measure to match the global single-item measure. For example, Ali *et al.*'s (2018a) review of PLS-SEM use in hospitality management reports an average sample size of 332 in relevant studies. In the context of our study, researchers would therefore need to measure the criterion construct with six or more indicators to produce the same conclusion that a global single item does. Given that many studies include multiple formatively measured constructs, using such a high number of items to validate each construct seems unreasonable.

	VAF values Reflective multi-item measure					
Sample size *N*	2	3	4	5	6	7
100	0.579	0.548	0.550	0.537	0.517	0.507
200	0.584	0.577	0.538	0.539	0.520	0.507
300	0.585	0.582	0.521	0.508	0.507	0.494
400	0.543	0.541	0.489	0.476	0.477	0.454
500	0.513	0.520	0.480	0.467	0.445	0.445

Table III. VAF values

Overall, these results illustrate that the convergent validity assessment of formatively measured constructs can be implemented without triggering a pronounced increase in survey length by simply using a global single item to measure the criterion construct. Our results show that the single item produces sufficient levels of convergent validity and excels its multi-item counterpart in terms of out-of-sample prediction. A potential reason for the latter finding is that the multi-item model overfits the original data by tapping spurious patterns in a sample because of its greater complexity (Myung, 2000).

Using a single item in PLS-SEM-based redundancy analyses is advantageous for researchers, who, to date, have largely ignored this important element of formative measurement model evaluation (Ali *et al.*, 2018a). However, researchers must consider two theoretical aspects when implementing a single-item-based redundancy analysis in their studies. First, the global single item must be specified in the research design phase *prior* to the data collection. Failure to do so will greatly complicate the convergent validity assessment, even rendering it impossible, unless secondary data are available that allow this type of analysis. However, this is generally not the case. Second, the global single-item measure must cover the essence of the construct it purports to measure. Answering questions about such a global single-item construct requires more abstract thinking from respondents than answering a set of multi-item measures, which typically have a high degree of semantical redundancy (Sloan *et al.*, 2002). Researchers should therefore offer a concise description of the concept under investigation to minimize the ambiguity when respondents endeavor to answer questions about each global single item. This specifically holds for heterogeneous concepts, such as the quality management in hotels (Pereira-Moliner *et al.*, 2016), the social capital of tourism firms (Martínez-Pérez *et al.*, 2016) and destination content gratification (Koo *et al.*, 2016), which prior hospitality management studies measured formatively.

While our results offer evidence for suitability of single items for redundancy analyses in the realm of our research context (i.e. service value), they do not necessarily generalize to other contexts, as different levels of average correlations within sets of indicators potentially have a strong impact on the model estimates (Hair *et al.*, 2017b; Sarstedt *et al.*, 2016c). Nevertheless, our results clearly show that if a multi-item measure of the criterion construct is highly homogeneous (as evidenced by high composite reliability values of around 0.9), a global single item that captures the essence of the construct's domain also explains a sufficient share of the multi-item measure's variance – independent of context. In case of doubt, researchers can administer single- and multi-item measures of the focal concept in a pilot study and run validation analyses, as described in our research. In case the single item yields sufficient levels of reliability and validity, researchers can administer only the single item in the main study, thereby still benefiting from its brevity.

Our study also provides the groundwork for establishing a validation procedure of single-item measures, which complements corresponding procedure multi-item measures (DeVellis, 2016). While our procedure considers fundamental validation steps that have been used in prior studies that empirically contrast single and multiple items, it is not without limitations. Future research should extend the analysis by, for example, test–retest reliability (Shamir and Kark, 2004) and discriminant validity (Gardner *et al.*, 1998).

On a final and more general note, our results illustrate the ambiguities that come with any redundancy analysis. While, for a sample size of 100, we would conclude that the formative measure of service value exhibits convergent validity, regardless of whether we use a global single item, or a multi-item measure as criterion construct, the same analysis would not support the convergent validity of larger sample sizes. Hence, the threshold values as reported by Chin (2010), Hair *et al.* (2017a, Chapter 5), Hsu *et al.* (2018) and

Ramayah *et al.* (2017) should only be considered rough rules of thumb. Rather than following a recipe-like, mechanistic approach to model evaluation, researchers should interpret the results in the context of the analysis – as recommended by methodological authorities such as Nunnally and Bernstein (1994), who also referred to the need to contextualize measurement (see also Rigdon *et al.*, 2017). Keeping these recommendations in mind, we expect redundancy analyses to become more prevalent in PLS-SEM studies that implement formative measurement models.

Notes

1. Researchers frequently distinguish between causal and composite indicators in formative measurement models. Causal indicators represent immediate causes of the focal construct whose error term captures all the other "causes" of the construct not included in the model (Diamantopoulos, 2006). Measurement models with composite indicators assume a definitional relation between indicators and constructs, in which the indicators explicitly define the construct's (empirical) meaning; its error term is therefore constrained to zero (Henseler, 2017). PLS-SEM treats indicators in formative measurement models as composite indicators (Hair *et al.*, 2017b). We use the term formative measurement model to refer to this type of model.
2. Note that other studies found that Cronbach's alpha is insensitive to sample size (Iacobucci and Duhachek, 2003).
3. Note that the first analysis corresponds to the squared multiple correlation (Wollenberg, 1977).
4. This conclusion only holds in the context of a redundancy analysis. Measuring a concept adequately still requires the use of multiple items (Sarstedt *et al.*, 2016a).

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Further reading

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Appendix

Service value index: confidence benefits (CB)	
I am convinced that the service at this fast food restaurant will be performed appropriately. (CB1)	
I have less anxiety when I buy/use the services of this fast-food restaurant. (CB2)	
I am less worried that something will go wrong at this fast food restaurant. (CB3)	
I know what to expect when I go to this fast-food restaurant. (CB4)	
I think I can trust this fast-food restaurant. (CB5)	
Service value index: perceived sacrifice (PS)	
The price charged for this fast-food restaurant's services is high. (PS1)	
The time required for this fast-food restaurant's services is long. (PS2)	
The effort I expect to receive this fast-food restaurant's services is high. (PS3)	
Service value index: service equity (SE)	
It makes more sense to dine in this fast-food restaurant than in others, even if they are the same. (SE1)	
Even if other fast-food restaurant offers the same service, I would still prefer this fast-food restaurant. (SE2)	
If other fast-food restaurant offers services as good as this fast-food restaurant, I would still prefer this one. (SE3)	
It seems wiser to dine in this restaurant even if the services of other fast-food restaurants are indifferent from this restaurant. (SE4)	
Service value index: service quality (SQ)	
In general, this fast-food restaurant's service is reliable and consistent. (SQ1)	
My experience with this fast-food restaurant is always excellent. (SQ2)	
I would say that this fast-food restaurant provides superior service. (SQ3)	
Overall, I think this fast food restaurant provides good service. (SQ4)	
Reflective multi-item criterion measure	
The value I received from this fast-food restaurant's services is worth the time, effort, and money that I invested. (SV1)	
This fast-food restaurant's services are reasonably priced. (SV2)	
This fast-food restaurant offers good services for the price. (SV3)	
I am happy with the price of this fast-food restaurant's services. (SV4)	
This fast-food restaurant makes me feel the money spent is worthwhile. (SV5)	
The value of this fast-food restaurant's services compares favorably to other others. (SV6)	
This fast-food restaurant offers good value for the price. (SV7)	
Global single-item criterion measure	
Overall, this fast-food restaurant provides a good service value. (SV8)	
Customer satisfaction	
I am happy with the service of this fast-food restaurant. (CS1)	
Overall, I am pleased when I dine in fast-food restaurant. (CS2)	
Using this fast-food restaurant's services is a satisfying experience. (CS3)	
My choice to dine in this fast-food restaurant is a wise one. (CS4)	
Overall, I am satisfied with this fast-food restaurant. (CS5)	
I think I did the right thing when deciding to dine in this fast-food restaurant. (CS6)	
Repurchase intention	
I will continue dining at this fast-food restaurant in the future. (RI1)	
As long as the present service continues, I doubt that I will switch fast-food restaurant. (RI2)	
I will choose this fast-food restaurant the next time I need this service. (RI3)	

Table AI.
Measurement items

Construct	Indicator	Mean	SD
Service quality	SQ1	3.919	0.733
	SQ2	3.741	0.814
	SQ3	3.624	0.771
	SQ4	3.877	0.748
Service equity	SE1	3.632	0.858
	SE2	3.541	0.976
	SE3	3.422	0.964
	SE4	3.370	0.918
Confidence benefits	CB1	3.840	0.709
	CB2	3.741	0.807
	CB3	3.640	0.882
	CB4	4.044	0.741
	CB5	3.968	0.747
Perceived sacrifice	PS1	2.903	1.027
	PS2	3.436	0.965
	PS3	3.582	0.795
Multiple item reflective measures of SV	SV1	3.752	0.753
	SV2	3.618	0.827
	SV3	3.661	0.762
	SV4	3.576	0.880
	SV5	3.600	0.876
	SV6	3.709	0.771
	SV7	3.747	0.752
Global single item of SV	SV8	3.861	0.723
Customer satisfaction	CS1	3.799	0.703
	CS2	3.804	0.704
	CS3	3.789	0.687
	CS4	3.706	0.758
	CS5	3.863	0.727
	CS6	3.770	0.763
Repurchase intention	RI1	3.981	0.771
	RI2	3.540	0.888
	RI3	3.853	0.749

Table AII.
Descriptive statistics

Table AIII.
Sample demographics

Demographic variable	Category	Frequency	(%)
Gender	Male	145	28.7
	Female	355	71.3
Ethnicity	Malay	162	32.4
	Chinese	260	52.0
	Indian	78	15.6
Education level	Degree	309	61.8
	Master	132	26.4
	PhD	59	11.8
Living arrangement	Stay with family	150	30.0
	Stay outside alone	350	70.0
Age	21-30 years old	360	72.0
	31-40 years old	100	20.0
	41-50 years old	40	8.0
Level of income	Less than RM2,000	275	55.0
	RM2,001-RM4,000	129	25.8
	RM4,001-RM6,000	81	16.2
	Above RM6,001	15	3.0

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